

EDUCATION

University of North Carolina at Chapel Hill | Kenan-Flagler Business School – Chapel Hill, NC May 2029

Bachelor of Science, Business Administration & Data Science | GPA: 4.0

- Honors: Assured Enrollment to Kenan-Flagler Business School; Honors Carolina: Top 10% of undergraduates
- Activities: Portfolio Management Team – TMT Sector Analyst | Quantitative Finance Association – Derivatives Analyst

The Lawrenceville School – Lawrenceville, NJ May 2025

High School Diploma | GPA: 3.92/4.0 UW

- Honors: Cum Laude Society; Herman Hollerith Prize Winner; Economics Club: NJ State Champion, National Finalist
- Publications: 2025 Journal of Future Economists (NY Federal Reserve): selected as top 12 of 195 papers nationwide

WORK EXPERIENCE

Cofounders Capital – Cary, NC Jan 2026 - Present

Venture Capital Intern

- Evaluated 100+ seed-stage B2B SaaS companies, analyzing market size, unit economics, competitive dynamics, and downside risk drivers to form investment views
- Synthesized diligence into reports assessing capital efficiency, risk–return tradeoffs, and exit viability for a \$60M fund

Hitech Corporation – Gaithersburg, MD / Mumbai, India July 2025 - September 2025

Marketing & Strategy Intern

- Designed U.S. market entry strategy via an automotive plastics acquisition by mapping supply chain tiers and identifying high-value OEM partners in the U.S. automotive ecosystem
- Built a CRM-integrated lead pipeline targeting \$50M+ TAM by profiling 20+ Tier-1 and aftermarket suppliers
- Analyzed competitor pricing, sourcing, and distribution models to design a 3-phase go-to-market (GTM) strategy

DTV.ai – Boston, MA May 2023 - February 2025

Co-founder

- Co-founded a B2B analytics firm modeling product unit economics to identify margin expansion opportunities
- Built and delivered 10+ cost breakdown reports identifying sourcing inefficiencies, design redundancies, and 12–15% improvement opportunities in per-unit economics
- Managed client relationships and negotiated contracts to generate \$40K+ in revenue from retail and CPG partners

PROJECTS

Working Paper (Quantitative Finance) | Research March 2025 - Present

Title: American Option Pricing Under Time-Varying Rough Volatility: A Signature-Based Hybrid Framework

- Published on arXiv (arxiv.org/abs/2508.07151v2) and SSRN; under review for journal publication
- Developed a hybrid American option pricing model integrating dynamic Hurst forecasting, regime-switching volatility engines, and fast signature-kernel approximations to improve accuracy and computational efficiency

High-Frequency Trading Order Book | Independent Project November 2025

Link to Repository: <https://github.com/roshanshah11/vertexladder>

- Developed a high-frequency trading (HFT) order book in C++ matching 1.87M+ orders/sec, applying financial markets engineering principles to simulate institutional execution workflows
- Optimized the matching engine to achieve $<0.41\mu\text{s}$ latency ($P99 \approx 0.67\mu\text{s}$), minimizing risk exposures and execution slippage for simulated market making strategies

VoiceBraille | M&TSI – University of Pennsylvania July 2024

Portable Speech-to-Braille Printer as part of Penn's M&TSI Program (~5% acceptance rate)

- Led a 4-person team through concept generation, CAD prototyping (SolidWorks), and financial modeling to design a portable speech-to-Braille printer
- Developed business and go-to-market plan targeting 200 schools and nonprofits; earned “Most Innovative Idea” Award

CAMPUS INVOLVEMENT

Kenan-Flagler Center for Decision Research September 2025 - Present

Research Assistant

- Apply behavioral science methods to 12 experimental studies on business decision-making in marketing and organizational behavior; working alongside PhD candidates under Professor Elad Sherf
- Execute focus groups, research surveys, and post-collection data cleaning for statistical analysis